

Instrumental Variables (IV)

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Motivation and Learning Objectives

So far, we have focused on methods like regression adjustment, matching, and weighting. All these methods share a common, crucial assumption: **ignorability**. They require us to believe that we have successfully measured and controlled for *all* common causes of the treatment and the outcome.

But what if we can't? What if there is a powerful, **unmeasured confounder** that we simply do not have in our dataset? This could be something like "patient motivation," "physician skill," or "innate ability." In these cases, our previous methods will fail to produce an unbiased causal estimate.

This week, we introduce **Instrumental Variables (IV)**, a brilliant and powerful method for estimating causal effects even in the presence of unmeasured confounding. The core idea is to find a "natural experiment" or a "random nudge" that affects the treatment but is not itself related to the outcome, except through its effect on the treatment.

By the end of this module, you will be able to:

1. **Identify** situations where unmeasured confounding makes standard methods inappropriate.
2. **Define** an instrumental variable and articulate its three core assumptions.
3. **Explain** the intuition behind IV estimation using the **Two-Stage Least Squares (2SLS)** method.
4. **Implement** IV estimation in R using the `ivreg` package.
5. **Understand** what the IV method actually estimates: the **Local Average Treatment Effect (LATE)**.
6. **Recognize and diagnose** the problem of "weak instruments."

Key Concepts: Hacking Causality with a "Nudge"

The Enduring Problem: Unmeasured Confounding

Let's consider a classic question: What is the causal effect of completing a college degree (D) on future earnings (Y)?

A simple regression of earnings on education will likely be biased. Why? Because individuals with higher **innate ability** (U) are both more likely to finish college and more likely to earn more, regardless of their education.

Since “ability” is nearly impossible to measure, it becomes an unmeasured confounder, creating a back-door path ($D \leftarrow U \rightarrow Y$) that we cannot close with standard methods.

The IV Solution: Finding a Clean Pathway

The IV approach seeks to find a variable—the **instrument** (Z)—that “nudges” people into treatment but is itself unrelated to the confounder.

An **instrumental variable** is a variable Z that affects the outcome Y *only* through its effect on the treatment D .

A classic example is using **geographic proximity to a college** as an instrument for education. People who live closer to a college (Z) may be more likely to attend (D), but where they live is probably not related to their innate ability (U).

The Three Core IV Assumptions

For a variable Z to be a valid instrument, it must satisfy three conditions:

1. **Relevance:** The instrument must have a causal effect on the treatment.
 - $Cov(Z, D) \neq 0$
 - *Intuition:* Proximity to a college must actually affect the probability that someone attends college. An irrelevant instrument is useless. We can and should test this with data.
2. **Exclusion Restriction:** The instrument must affect the outcome *only* through the treatment. It cannot have a direct effect on the outcome.
 - *Intuition:* Living near a college cannot *directly* increase your earnings. It can only do so by making you more likely to get a degree. This is a strong assumption and cannot be proven with data; it must be justified by theory and domain knowledge.
3. **Exchangeability (or Independence):** The instrument must not share any common causes with the outcome. It must be “as-if” randomly assigned with respect to the unmeasured confounders.
 - $Cov(Z, U) = 0$
 - *Intuition:* The types of people who live near a college cannot be systematically different (in terms of ability) from those who live far away.

💡 Justifying the IV Assumptions is Key

The validity of an IV analysis rests entirely on the plausibility of these three assumptions. While relevance can be tested, the exclusion restriction and exchangeability are untestable assumptions that you must defend based on institutional knowledge and logical argument.

Numerical Example: Two-Stage Least Squares (2SLS)

The most common IV estimator is **Two-Stage Least Squares (2SLS)**. The intuition is beautiful: we “cleanse” the endogenous treatment variable of its correlation with the unmeasured confounder.

Imagine our goal is to estimate: $Y = \beta_0 + \beta_1 D + \epsilon$

But we know D is correlated with ϵ (because of U).

Stage 1: Isolate the “Clean” Variation in Treatment First, we predict the treatment D using only our instrument Z .

$$D = \alpha_0 + \alpha_1 Z + \nu$$

The predicted values from this regression, $\hat{D}$, represent the part of the treatment that is “exogenous”—i.e., the part that is purely driven by our “as-if” random instrument.

Stage 2: Estimate the Causal Effect Second, we regress the outcome Y on the *predicted* treatment from Stage 1.

$$Y = \beta_0 + \beta_1 \hat{D} + \epsilon'$$

Because $\hat{D}$ is, by construction, uncorrelated with the unmeasured confounder U , the resulting estimate of β_1 is our unbiased IV estimate of the causal effect.

A Worked Example in R

Let’s simulate data for the education-earnings example. We’ll set the true causal effect of education to 5 (e.g., \$5,000).

```

# Load necessary libraries
library(dplyr)
library(ivreg)
library(AER) # Contains the dataset we'll use

# --- 1. Simulate data with an unmeasured confounder ---
set.seed(123)
n <- 1000
# Z: Instrument (Proximity to college, 1=close, 0=far)
# U: Unmeasured confounder (Ability)
# D: Treatment (Gets a degree)
# Y: Outcome (Earnings)
data <- tibble(
  Z = rbinom(n, 1, 0.5),
  U = rnorm(n, 0, 1),
  # Treatment depends on the instrument Z and ability U
  D = rbinom(n, 1, pnorm(0.5 * Z + 1.5 * U - 0.5)),
  # Outcome depends on treatment D and ability U. True effect of D is 5.
  Y = 10 + 5 * D + 3 * U + rnorm(n, 0, 2)
)

# --- 2. OLS (Naive) Regression ---
# This is biased by the unmeasured confounder U
ols_model <- lm(Y ~ D, data = data)
summary(ols_model)
# The coefficient on D is ~8.1, which is badly biased upwards from the true value of 5.

# --- 3. IV Regression using 2SLS ---
# The syntax for ivreg is Y ~ treatment | instrument(s)
# It performs both stages for us automatically.
iv_model <- ivreg(Y ~ D | Z, data = data)
summary(iv_model)
# The coefficient on D is now ~4.9, very close to our true effect of 5!

```

IV in the Potential Outcomes Framework: What is the LATE?

The IV estimate is not the Average Treatment Effect (ATE). Instead, it is the **Local Average Treatment Effect (LATE)**. To understand this, we need to define four types of people in our population relative to the instrument:

1. **Compliers:** These are individuals who take the treatment *if* encouraged by the instrument, and don't if they are not. (They comply with the “nudge”).
2. **Always-Takers:** These individuals will take the treatment regardless of the instrument's value.
3. **Never-Takers:** These individuals will *never* take the treatment, regardless of the instrument.
4. **Defiers:** These individuals do the opposite of what the instrument encourages. (This group is generally assumed not to exist).

The LATE theorem (Angrist & Imbens, 1994) states that the IV estimate is the average treatment effect *only for the compliers*.

Concept Check

Let's use a military draft lottery as an instrument for veteran status to study the effect of military service on health. Who are the “compliers”?

Answer: The compliers are the men who would only serve in the military if they were drafted, but would not volunteer otherwise. The IV estimate gives us the effect of military service *only for this specific subgroup*, not for the always-takers (volunteers) or never-takers (those who would avoid service even if drafted).

The Problem of Weak Instruments

A “weak instrument” is an instrument that satisfies the exclusion and exchangeability assumptions, but is only weakly correlated with the treatment (the **Relevance** assumption is weak).

- **Consequence 1:** The IV estimate becomes biased in the direction of the OLS estimate.
- **Consequence 2:** The standard errors of the IV estimate become very large.

Rule of Thumb: In the first stage regression ($D \sim Z$), check the F-statistic for the significance of the instrument. If the **F-statistic is less than 10**, you likely have a weak instrument problem.

Appendix: Advanced Statistical Derivations and Theory

Derivation of the Simple IV Estimator

Motivation: This derivation shows how the ratio of two covariances can produce a consistent estimator for the causal effect β_1 in the presence of an unmeasured confounder.

The Model: Assume the true structural model is:

1. $Y = \beta_0 + \beta_1 D + U$ (where U is the unmeasured confounder)
2. $D = \alpha_0 + \alpha_1 Z + V$ (where Z is the instrument)

We assume the key IV property: $Cov(Z, U) = 0$. Our goal is to estimate β_1 .

1. **Start with the covariance between the outcome Y and the instrument Z :**

$$Cov(Y, Z) = Cov(\beta_0 + \beta_1 D + U, Z)$$

2. **Using the properties of covariance, this expands to:**

$$Cov(Y, Z) = Cov(\beta_0, Z) + \beta_1 Cov(D, Z) + Cov(U, Z)$$

2. **Simplify the terms:**

- $Cov(\beta_0, Z) = 0$ (covariance with a constant is zero).
- $Cov(U, Z) = 0$ (this is our crucial **exchangeability** assumption).

3. **This leaves us with a simple expression:**

$$Cov(Y, Z) = \beta_1 Cov(D, Z)$$

5. **Solve for our target parameter, β_1 :**

$$\beta_1 = \frac{Cov(Y, Z)}{Cov(D, Z)}$$

Conclusion: The true causal effect β_1 is the ratio of the “reduced form” effect of the instrument on the outcome (the numerator) to the “first stage” effect of the instrument on the treatment (the denominator). The sample counterparts of these covariances form the simple IV estimator:

$$\hat{\beta}_{IV} = \frac{\sum (Y_i - \bar{Y})(Z_i - \bar{Z})}{\sum (D_i - \bar{D})(Z_i - \bar{Z})}$$